ELECTRONIC SPECTROSCOPY OF ATOMS

□ State and explain building up principles with example.

Ans:

There are three basic rules known as the builing up rules, which determine how electrons in large atoms occupy orbitals.

• Pauli's Principle: No two electrons in an atom may have the same set of values for $n, l, l_z (m)$ and s_z.

According to this principle, if any atom contains two electron in the same orbital, then the value of n, l and $l_z (m)$ must be the same but their spin must be different. The four quantum numbers of two electrons can never be identical. He contains two electron in the 1s orbital.

For the first electron, $n=1$, $l=0$, $m=0$, $s=+1/2$

" " " " , $n=1$ $l=0$, $m=0$, s must be $=-1/2$.

• Aufbau principle: Electrons tend to occupy the orbital with lowest energy available.

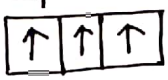
According to the rule, electrons first occupy the lowest possible energy levels and then gradually move to the higher energy levels. Orbital energies depend on the $(n+l)$ value. With the increasing value of n and l, orbital energies also increase. Electrons occupy energy levels according to the following order:

$1S < 2S < 2P < 3S < 3P < 4S < 3d < 4P < 5S - - - -$

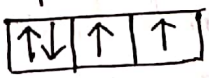
Scanned with OKEN Scanner

• Hund's Principle: Electrons tend to occupy degenerate orbitals singly with their spins parallel. When all the degenerate orbitals are filled with single electron having parallel spin, then other electron would move either of the orbitals being paired.

N: $1s^2 2s^2 2p^3$



O: $1s^2 2s^2 2p^4$



□ What are meant by hydrogen and helium-like species? Give examples.

Ans:

Hydrogen has only one electron in it's 1s orbital. Any atom or ion which contains a single electron in it's valance shell is called Hydrogen like species. Example:

He^+ (2): $1s^1$

Be^+ (4): $1s^2 2s^1$

Li (3): $1s^2 2s^1$

The above species have one electron in the outer shell and their spectra will be same as hydrogen atom.

Helium has two electrons in it's valance shell. Thus any atom or ion containing two valance electron is known as He like species. Example:

Scanned with OKEN Scanner

He (2) : $1s^2$

Be^{2+} (4) : $1s^2$

B^+ (5) : $1s^2\ 2s^2$

C^{2+} (6) : $1s^2\ 2s^2$

All the mentioned species have 2 electrons in its outer shell and produce spectra same as He atom.

□ Describe the transition from ^2P to ^2D states for H atom spectrum with necessary energy diagram and spectrum. Why is the spectrum obtained from $^2P \rightarrow {}^2D$ transitions reffered to as compound doublet?

Ans8

The selection rule for transition of electron in hydrogen atom,

Δn = anything $\Delta l = \pm 1$ and $\Delta J = 0, \pm 1$

These selection rules indicate that the transition between ^2P and ^2D levels are quite more complex.

For p orbital, $l = 1$; thus doublet with, $J = 1 + 1/2 = 3/2$ and $J = 1 - 1/2 = 1/2$ occurs.

Again, for d orbital, $l = 2$; thus doublet with, $J = 2 + 1/2 = 5/2$ and $J = 2 - 1/2 = 3/2$ occurs.

Scanned with OKEN Scanner

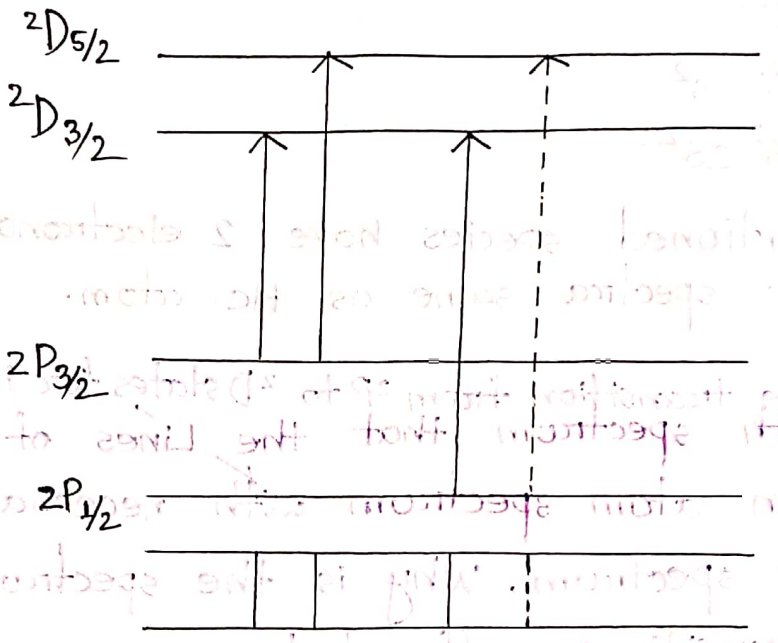


Fig: The 'compound doublet' spectrum arraising as a result of transitions between ^2P and ^2D levels in the H atom.

Transitions between ^2P and ^2D involves 4 energy levels. Transitions between $^2P_{3/2}$ and $^2D_{3/2}$ corresponds to $\Delta J = 0$ which has the lowest frequency. Thus the transition is allowed. Transitions from $^2P_{3/2} \longrightarrow {}^2D_{5/2}$ and $^2P_{1/2} \longrightarrow {}^2D_{3/2}$ is allowed as for both the cases $\Delta J = \pm 1$. The fourth transition $^2P_{1/2} \rightarrow {}^2D_{5/2}$ is not allowed since $\Delta J = +2$.

As, transition from ^2P to ^2D energy levels in hydrogen atom consists of three spectral lines arising ~~from~~ between doublet levels, it is referred as 'doublet compound' spectrum.

Scanned with OKEN Scanner

□ Show with spectrum that the lines of Lymen series of hydrogen atom are observed at $\frac{3R}{4}, \frac{8R}{9}, \frac{15R}{16}$ ---- cm^{-1} in the spectrum.

Ans:

The electron configuration of H -atom,

$H\ (1) : 1s^1$

The selection rule for electronic transition of H,

Δn = anything

$\Delta l = \pm 1$

Thus, the electron in the ground state can undergo a transition into any p state:

$$1s \longrightarrow np \quad (n \geq 2)$$

In general, the electron in a lower state n'' can undergo a higher state n' with the absorption of energy,

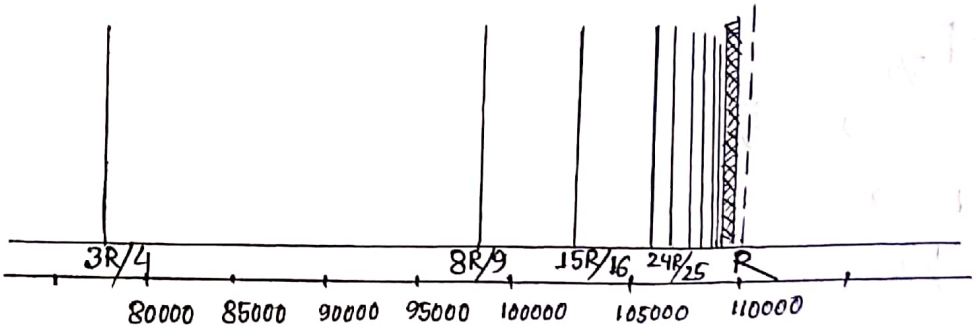
$$\bar{\upsilon}_{spect} = R\left(\frac{1}{n''^2} - \frac{1}{n'^2}\right) cm^{-1}$$

For transition $1s \rightarrow np$, lines of spectrum occurs are known as Lymen series,

$$\bar{\upsilon}_{Lymen} = R\left(\frac{1}{1^2} - \frac{1}{n'^2}\right) cm^{-1} \text{ where, } (n' = 2, 3, 4 \cdots)$$

putting $n' = 2, 3, 4 \cdots$ respectively,

$$\bar{\upsilon}_{Lymen} = \frac{3R}{4}, \frac{8R}{9}, \frac{15R}{16} \cdots\cdots cm^{-1}$$



Scanned with OKEN Scanner

□ Find the values of L, S and J for:

(i) $^4S_1$ (ii) $^2D_{7/2}$ (iii) $^4P_{3/2}$ (iv) $^2D_{5/2}$ (v) $^3S_1$

Ans:

(i) $^4S_1$

Term symbol $= {}^{2S+1}L_J$

Here, $J = 1$, $L = 0$ (as $S = 0$)

And, $2S + 1 = 4$

$\Rightarrow 2S = 3$

$\therefore S = 3/2$

(ii) $^2D_{7/2}$

$L = 2$; $J = 7/2$,

& $2S + 1 = 2$

$\Rightarrow 2S = 1$

$\therefore S = 1/2$

(iii) $^4P_{3/2}$

$L = 1$, $J = 3/2$

$\Rightarrow 2S + 1 = 4$

$\Rightarrow 2S = 3$

$\therefore S = 3/2$

(iv) $^2D_{5/2}$

$L = 2$; $J = 5/2$

$2S + 1 = 2$

$\therefore S = 1/2$

(v) $^3S_1$

$L = 0$; $J = 1$

$2S + 1 = 3$

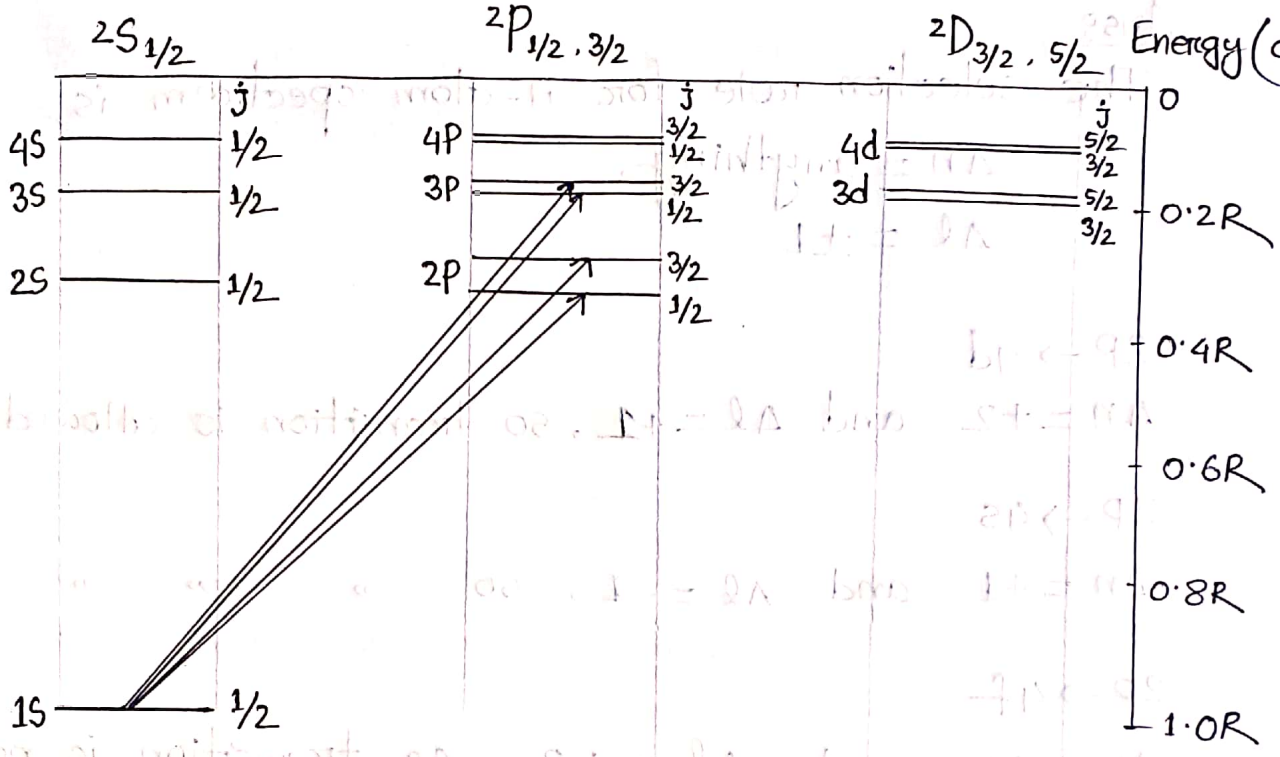
$\Rightarrow 2S = 2$

$\therefore S = 1$

Scanned with OKEN Scanner

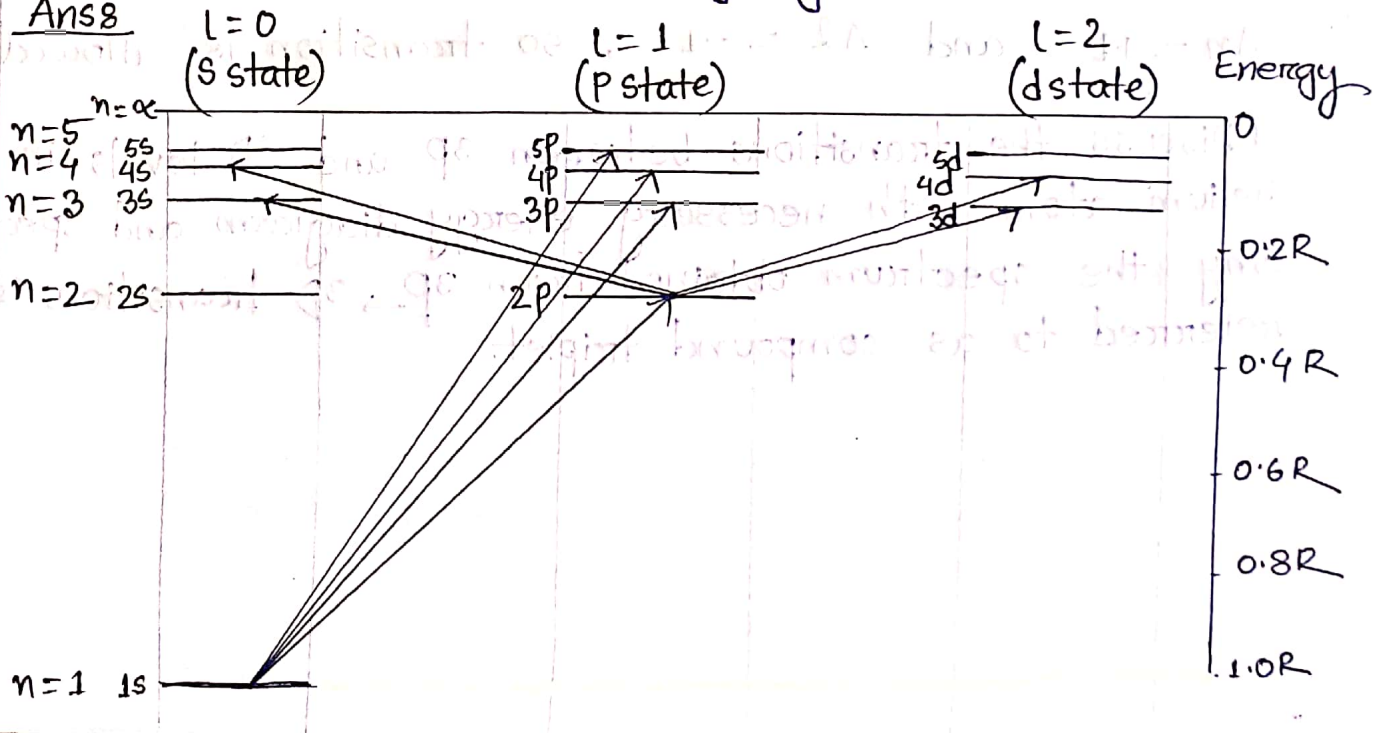
□ Sketch the electronic energy levels and show transition between them for single electron of hydrogen atom. showing inclusion of j-splitting.

Ans 8



□ Sketch the electronic energy levels and show transition between them for single electron of hydrogen atom.

Ans 8



Scanned with OKEN Scanner

□ State the selection rule for H-atom spectrum. Which of the following transitions are allowed for H-atom spectrum ?

$2P \rightarrow 4d$, $3P \rightarrow 4S$, $2P \rightarrow 4f$, $2P \rightarrow 1S$

Ans:

The selection rule for H-atom spectrum is,

$\Delta n =$ anything

$\Delta l = \pm 1$

$2P \rightarrow 4d$

$\Delta n = +2$ and $\Delta l = +1$, so transition is allowed.

$3P \rightarrow 4S$.

$\Delta n = +1$ and $\Delta l = -1$, so " " "

$2P \rightarrow 4f$

$\Delta n = +2$ and $\Delta l = +2$, so transition is not allowed.

$2P \rightarrow 1S$

$\Delta n = +1$ and $\Delta l = -1$, so transition is allowed.

□ Discuss the transitions between 3P and ^3D levels in the helium atom with necessary energy diagram and spectrum. Why the spectrum obtained from $3P \rightarrow {}^3D$ transitions is referred to as compound triplet.

Scanned with OKEN Scanner

Answer:

Transition from the ^3P states of He may take place either to ^3S or ^3D states. Since, the selection rule is,

$$\Delta S = 0 \quad , \quad \Delta L = \pm 1, \quad \Delta J = 0, \pm 1$$

Transition from ^3P to ^3D; the spectral series may be very complex if completely resolved.

For ^3P we have,

$$S = 1 \quad , \; L = 1 \quad , \; J = 2, 1, 0$$

So the term symbol $\rightarrow$ $^3P_2$, $^3P_1$, $^3P_0$

For ^3D we have;

$$S = 1 \quad , \; L = 2 \quad J = 3, 2, 1$$

Term symbol $\rightarrow$ $^3D_3$, $^3D_2$, $^3D_1$

According to the selection the transitions can be shown as:

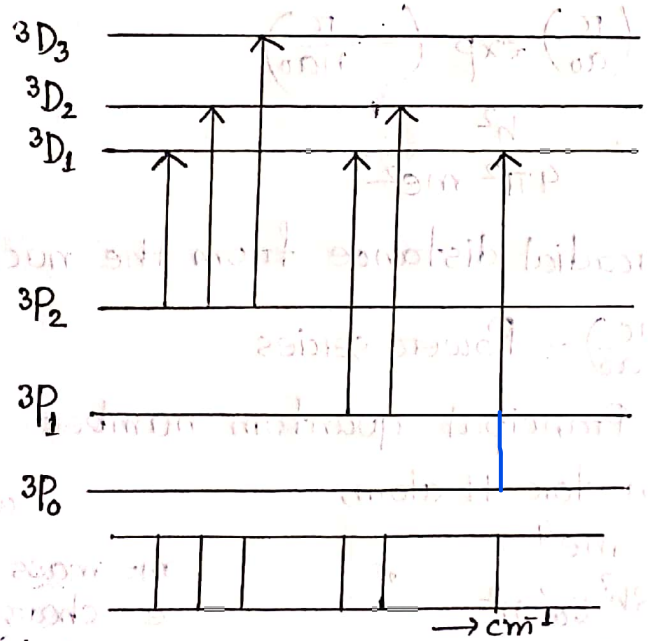


Fig: Compound Triplet spectrum arising from transitions between ^3P and ^3D levels in the helium atom.

Scanned with OKEN Scanner

Transitions between $^3P_2$ and $^3D_3$, $^3D_2$, $^3D_1$ corresponds to $\Delta J = +1, 0, -1$ respectively. Transition from $^3P_1$ allows only to $^3D_1$ and $^3D_2$. Perhaps only one transition is possible from $^3P_0$ to $^3P_1$. Thus, the complete spectrum should consist of six lines. Normally, however, the very close spacing is not resolved, and only three lines are seen; for this reason the spectrum is referred to as a compound triplet.

□ Write down the wave function expression and energy expression for H atom.

Ans:

The wave function expression for H atom,

$$\psi_n = f\left(\frac{r}{a_0}\right) \exp\left(-\frac{r}{na_0}\right)$$

where, $a_0 = \dfrac{h^2}{4\pi^2 me^2}$

r = radial distance from the nucleus

$f\left(\frac{r}{a_0}\right)$ = Power series

n = Principal quantum number.

Energy expression for H atom,

$$E_n = -\frac{me^4}{8h^2\varepsilon_0^2 n^2} \text{ J}$$

$$\varepsilon_n = -\frac{me^4}{8h^3 c\, \varepsilon_0^2 n^2} \text{ cm}^{-1}$$

$$= -\frac{R}{n^2} \text{ cm}^{-1} \quad (n = 1, 2, 3 \ldots)$$

where,

m = mass of electron
e = charge of electron
h = Plank's constant
ε_0 = Vacuum permittivity
R = Rydberg constant
n = Principal quantum number.

Scanned with OKEN Scanner

□ What are atomic quantum numbers? Discuss their significance

Ans:

The set of numbers used to describe the position and energy of the electron in an atom are called atomic quantum numbers, namely principal, azimuthal, magnetic and spin quantum numbers.

The significance of the four types of atomic quantum numbers is given below :-

Quantum Number	Allowed values	Function
Principal Quantum number, n	$n = 1, 2, 3 \ldots$	Governs the size and energy of the orbital.
Azimuthal quantum number, l	$l = (n-1), (n-2), \ldots 0$	Governs the shape of the orbital and electronic angular momentum.
Magnetic quantum number, m	$m = \pm l, \pm (l-1) \ldots 0$	Governs the direction of an orbital and electron behaviour in a magnetic field.
Spin quantum number, s	$s = \pm 1/2$	Governs axial angular momentum of the electron.

Scanned with OKEN Scanner

□ Sketch the energy levels of lithium atom in a figure showing the difference in energy of s, p and d states with the same value of n and also show some allowed transitions in the figure.

Ans:

The electronic configuration of Li is,

$$Li(3) \rightarrow 1s^2\ 2s^1$$

The alkali metal like Li, Na, K, Rb all have single electron in the outer shell. Hence they show hydrogen like spectra. The selection rules for alkali metal is same as the H atom; i.e.

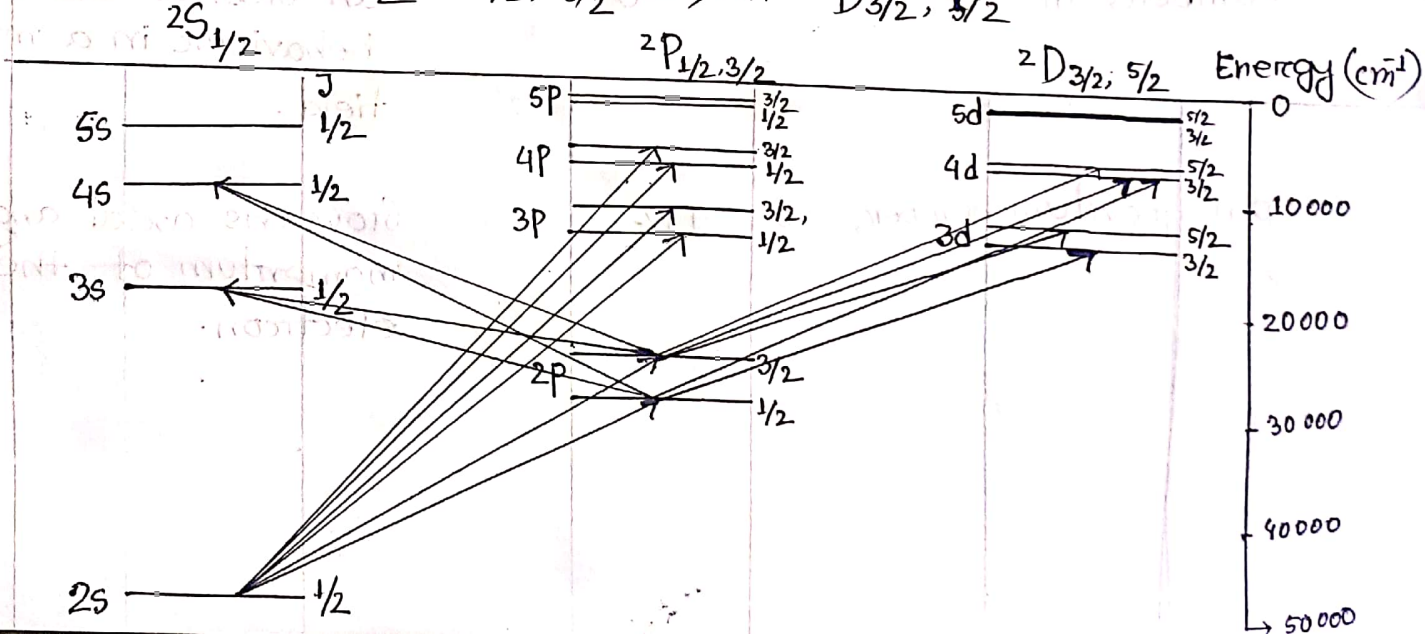
$$\Delta n = \text{anything}\ ,\ \Delta l = \pm 1\ ,\ \Delta J = 0, \pm 1$$

Thus transitions from the ground state $(1s^2 2s)$ can occure to p levels: $^2S_{1/2} \longrightarrow n\,^2P_{1/3,\,3/2}$ and a series of doublets similiare to the Lyman series will be formed.

From 2P states, two seperate series of lines will be seen:

$$2\ ^2P_{1/2,\,3/2} \longrightarrow n\ ^2S_{1/2}$$

$$2\ P_{1/2,\,3/2} \longrightarrow n\ ^2D_{3/2,\,5/2}$$



Scanned with OKEN Scanner

□ What is the selection rule for many electron system in case of electronic spectroscopy?

Ans:

The selection rules for many electron systems are:

$$\Delta S = 0, \quad \Delta L = \pm 1 \quad \Delta J = 0, \pm 1$$

□ How can you get the term symbol for the following?

(i) $L = 1$, $S = 1$ and $J = 2$

(ii) $L = 2$, $S = 1/2$ and $J = 3/2$

Ans:

(i) Term symbol $= {}^{2S+1}L_J$

Here, $2S+1 = 2*1+1 = 3$

$L = 1 = P$

$J = 2$

$\therefore$ Term symbol $= {}^3P_2$

(ii) $2S+1 = 2 \times 1/2 + 1 = 2$

$L = 2 = D$, $J = 3/2$

$\therefore$ Term symbol $= {}^2D_{3/2}$

Scanned with OKEN Scanner

□ Which electronic transition can give rise to Lyman and Balmer spectral lines? Mention their absorption energy.

Ans:

The Schrodinger equation shows the energy for H-atom:

$$E = -\frac{me^4}{8h^2\pi^2n^2} \text{ J}$$

$$\varepsilon = -\frac{me^4}{8h^2c\pi^2n^2} \text{ cm}^{-1} \quad (n = 1, 2, 3\cdots)$$

$$= -\frac{R}{n^2} \text{ cm}^{-1}; \text{ where, } R = -\frac{me^4}{8h^2c\pi^2}$$

When an electron in lower state n'' undergo a transition into a higher state n', the absorption of energy,

$$\Delta\varepsilon = \varepsilon_{n'} - \varepsilon_{n''}$$

$$= -\frac{R}{n'^2} + \frac{R}{n''^2} = R\left(\frac{1}{n''^2} - \frac{1}{n'^2}\right) \text{cm}^{-1}$$

Transitions from $1s \longrightarrow n'p$ in H atom, causes a series of spectrum to appear named Lyman series.

Transitions from $2s \longrightarrow n'p$ or $2p \rightarrow n's, n'd$ cause a series of Balmer spectral lines to appear.

Absorption energy for Lyman series where $n'' = 1$,

$$\bar{\upsilon}_{Lyman} = \Delta\varepsilon = R\left(1 - \frac{1}{n'^2}\right) \text{cm}^{-1}$$

$$= \frac{3R}{4}, \frac{8R}{9}, \frac{15R}{16} \cdots \text{cm}^{-1} \quad (n' = 2, 3, 4\cdots)$$

Scanned with OKEN Scanner

Absorption energy for Balmer series, where, $n'' = 2$,

$$\bar{\upsilon}_{Balmer} = R\left(\frac{1}{2^2} - \frac{1}{n'^2}\right) cm^{-1}$$

$$= \frac{5R}{36}, \frac{3R}{16}, \frac{21R}{100} \cdots cm^{-1} \quad (n' = 3, 4, 5 \cdots)$$

□ What are the factors that govern the fine structure of the hydrogen atom spectrum?

Ans:

The factors that govern the fine structure of H spectrum are :-

(i) Orbital Angular momentum: An electron moving in its own orbital about a nucleus possesses orbital angular momentum.

Orbital Angular momentum, $l = \sqrt{l(l+1)}\ h/2\pi$

(ii) Spin Angular momentum: Every electron in atom spinning about an axis possesses spin angular momentum.

Spin angular momentum, $s = \sqrt{s(s+1)}\ \frac{h}{2\pi}$

(iii) Total Angular momentum: Coupling of both orbital and spin angular momentum causes the fine structure of H-atom to be splitted.

$$j = \sqrt{j(j+1)}\ \frac{h}{2\pi}, \quad j = l + s$$

Scanned with OKEN Scanner